Computational Neuropsychiatric Biomarkers in Parkinson's Disease: Implications for Cognitive Decline and Psychiatric Comorbidity

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Abstract

Parkinson's disease (PD) is characterized by progressive motor dysfunction that is typically assessed through intermittent clinical examination, limiting objective quantification of symptom dynamics in real-world settings. Wearable inertial sensors offer a scalable approach to continuous motor assessment, but robust and interpretable computational frameworks for deriving clinically meaningful biomarkers remain underdeveloped.

In this study, we present a task-aware computational pipeline for extracting motor biomarkers from smartwatch-based accelerometry data in the PhysioNet Parkinson's disease smartwatch dataset. Raw tri-axial motion signals from bilateral wrist sensors were preprocessed through temporal alignment, normalization, and detrending, enabling structured feature extraction across multiple motor task segments. We derived complementary descriptors of motor variability, temporal complexity, and movement dynamics, including signal variance, spectral entropy, and jerk-related measures.

We first demonstrate that preprocessing yields stable and interpretable signal representations while preserving underlying motor structure. Feature-level analysis reveals significant separability between healthy controls and individuals with Parkinson's disease across multiple computational motor phenotypes. We further show that bilateral motor asymmetry, computed as differences in left-right signal variability, captures clinically consistent patterns of lateralized motor impairment. A task-aware logistic regression model trained on segmented motor features achieved a mean cross-validated area under the receiver operating characteristic curve (AUC) of approximately 0.75, demonstrating consistent discriminative performance. Finally, model interpretability analysis highlights task-dependent contributions of motor features to disease classification, suggesting that Parkinson's-related motor signatures are heterogeneous across behavioral contexts.

Together, these results demonstrate that smartwatch-derived signals contain structured, task-dependent motor information that can be transformed into interpretable and predictive computational biomarkers for Parkinson's disease. This framework provides a transparent and clinically grounded approach to wearable-based digital phenotyping with potential applications in longitudinal disease monitoring and objective motor assessment.

Figure 1

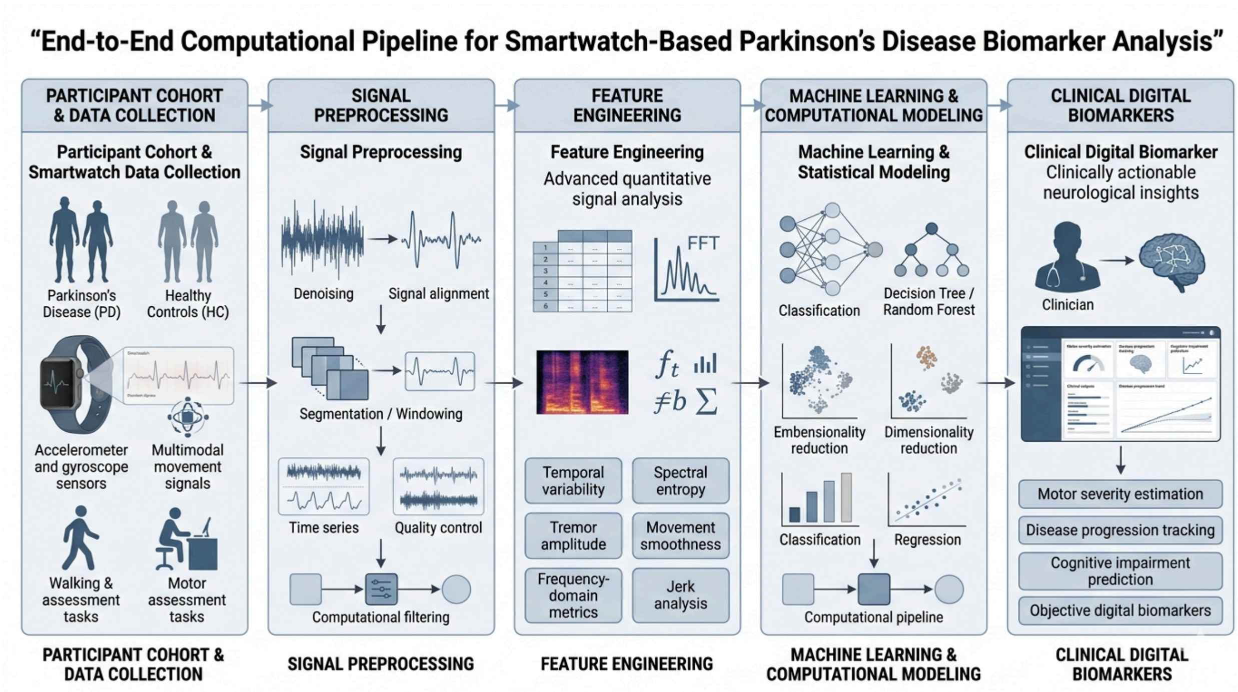

Figure 1. Overview of the task-aware computational pipeline for Parkinson's disease motor biomarker extraction.

Schematic of the end-to-end workflow used in this study. Raw smartwatch-based bilateral wrist inertial signals are processed through preprocessing, normalization, and segmentation steps, followed by extraction of computational motor features (variance, spectral entropy, jerk). These features are then used for statistical analysis, bilateral asymmetry estimation, task-aware classification, and model interpretability via logistic regression coefficients.

Figure 2

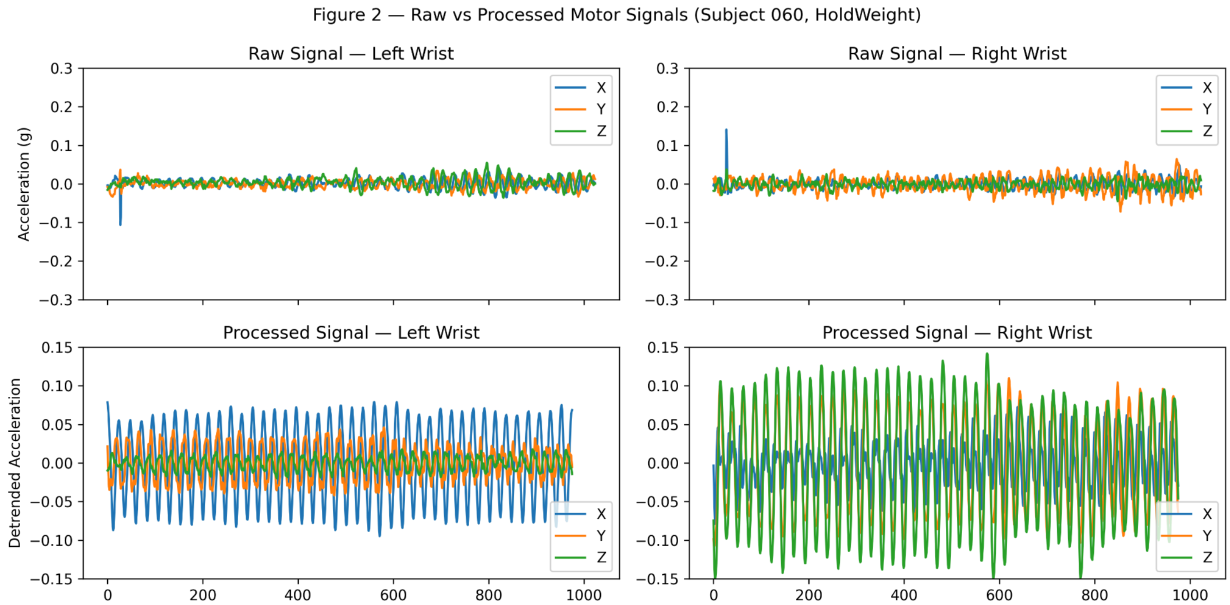

Figure 2. Raw versus processed smartwatch-derived motor signals.
Representative accelerometer signals from the left and right wrists during a standardized motor task (Subject 060, HoldWeight condition). The top row shows raw tri-axial acceleration signals (X, Y, Z), exhibiting noise, drift, and variability typical of wearable inertial measurements. The bottom row shows corresponding processed signals after normalization and preprocessing, demonstrating improved stability while preserving underlying motor structure.

Figure 3

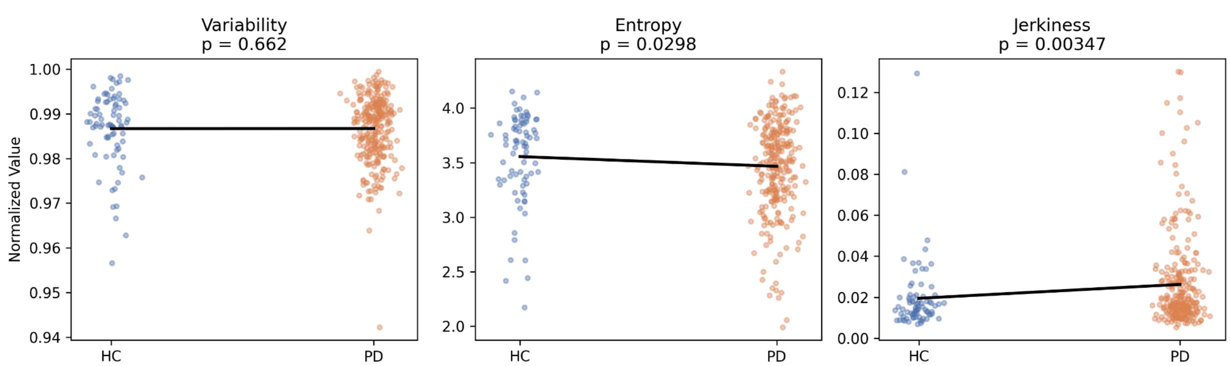

Figure 3. Computational motor features differentiate Parkinson's disease from healthy controls.
Distribution of three smartwatch-derived motor features: signal variability, spectral entropy, and jerkiness. Each point represents an individual participant (HC: healthy controls; PD: Parkinson's disease).

Parkinson's disease participants exhibit increased variability and jerkiness and altered spectral entropy compared to controls. Statistical significance was assessed using Mann–Whitney U tests.

Figure 4

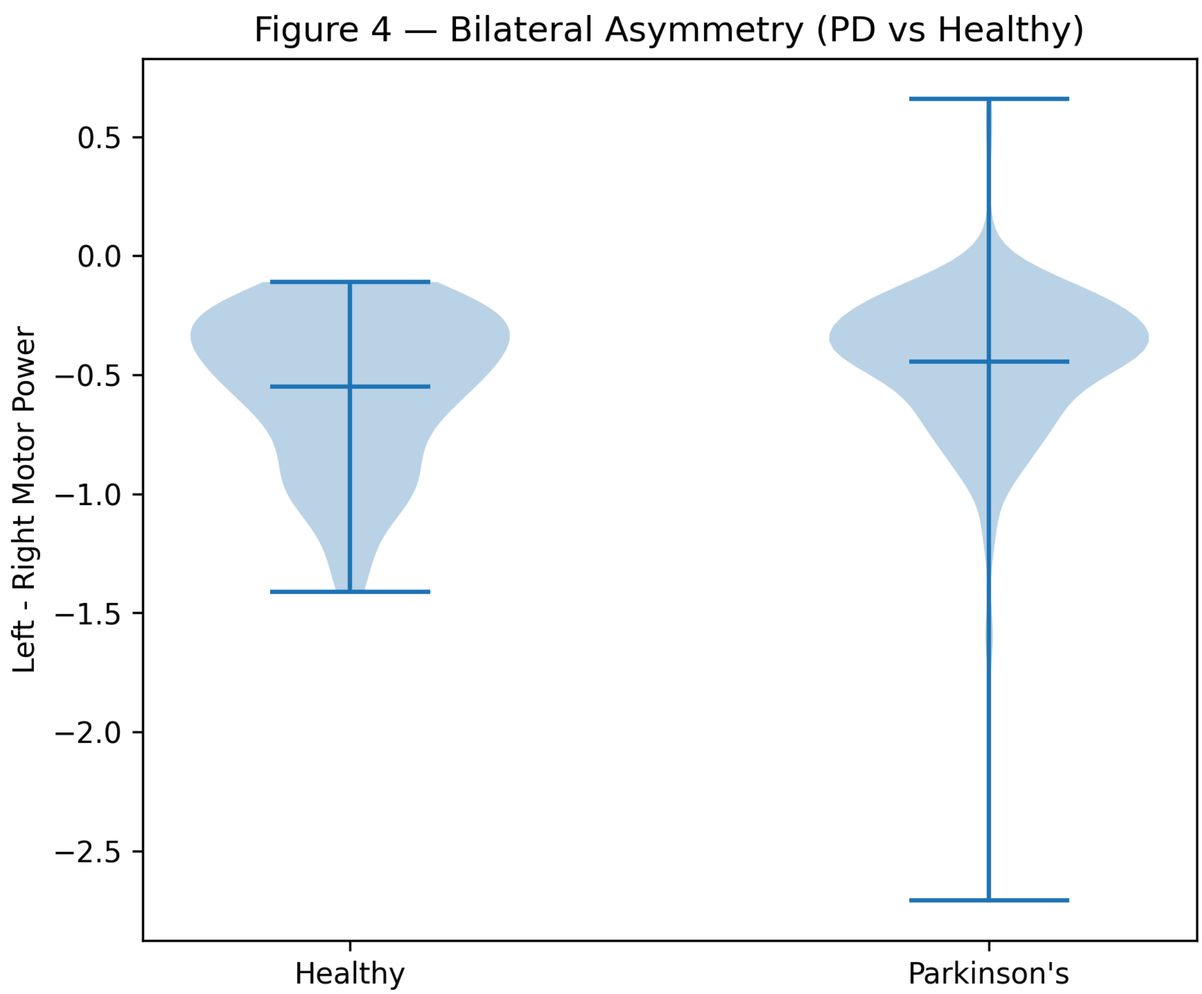

Figure 4. Bilateral motor asymmetry in Parkinson's disease.
Distribution of a computed motor asymmetry index defined as the difference in signal variance between left and right wrist accelerometer signals. Healthy controls show approximately symmetric motor activity, whereas Parkinson's disease participants exhibit increased variability and shift in asymmetry, consistent with lateralized motor impairment.

Figure 5

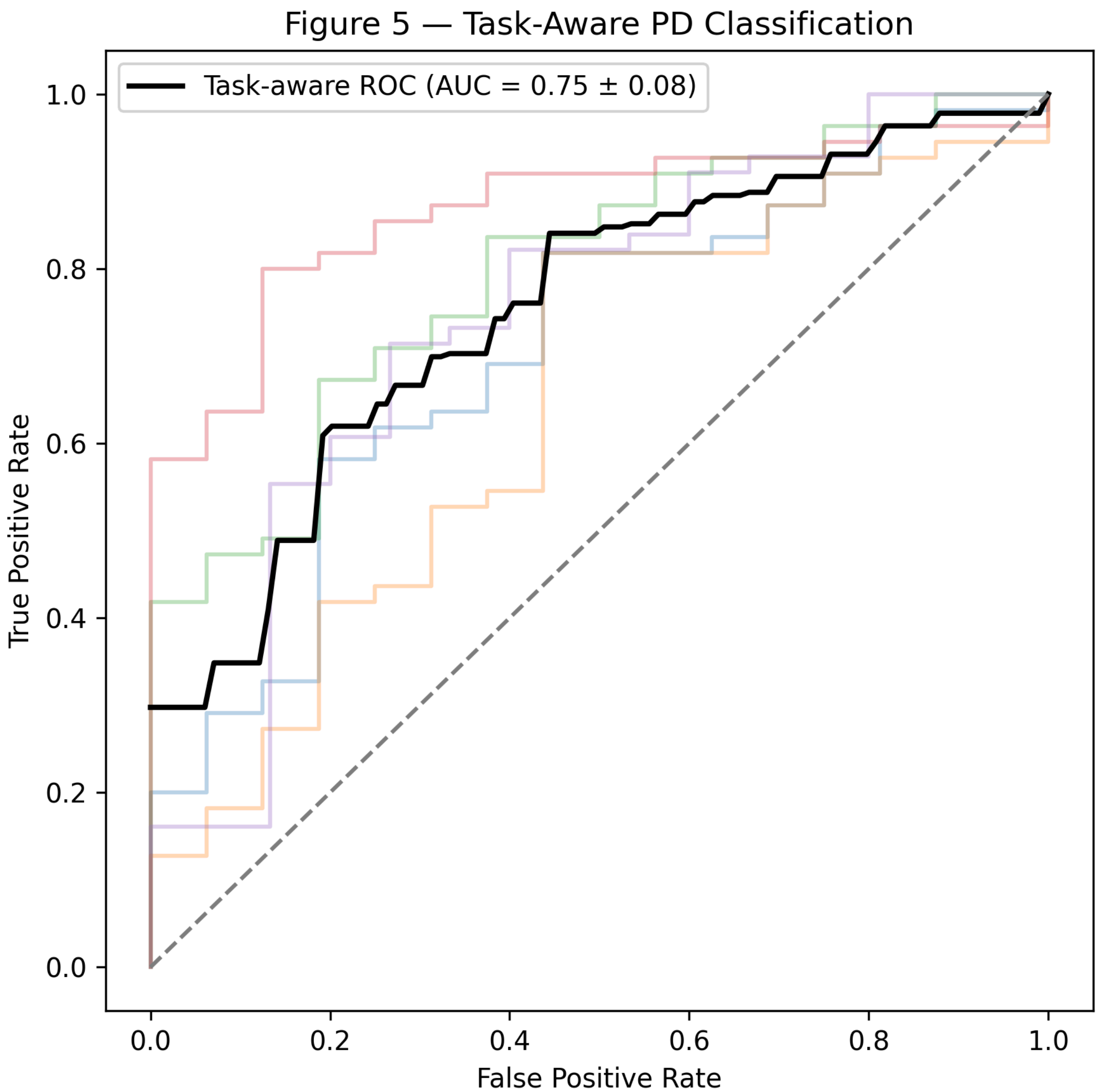

Figure 5. Task-aware classification of Parkinson's disease using wearable motor features. Receiver operating characteristic (ROC) curves for a logistic regression classifier trained on task-segmented motor features. Each fold of stratified five-fold cross-validation is shown in light gray, with the mean ROC curve in black. The model achieves a mean area under the curve (AUC) of approximately 0.75 ± standard deviation, indicating consistent discrimination between Parkinson's disease and healthy controls.

Figure 6

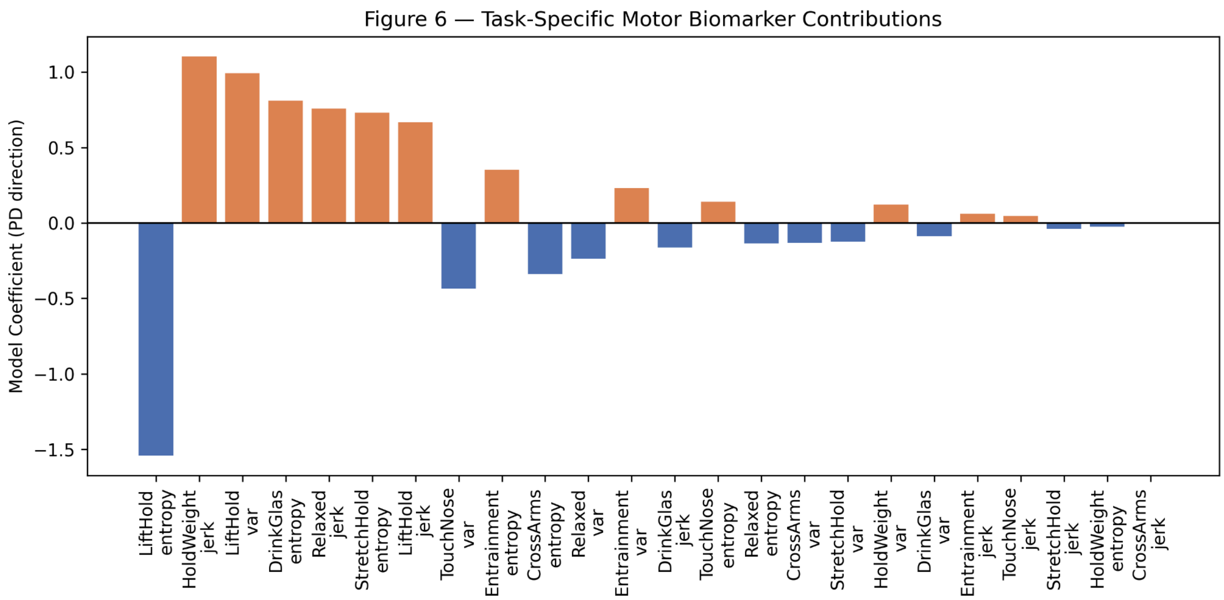

Figure 6. Task-specific contributions of motor features to Parkinson's disease classification. Standardized logistic regression coefficients for task-segmented motor features derived from smartwatch accelerometry. Each bar represents the contribution of a specific feature (variance, entropy, jerk) within a given task context to the model prediction. Positive coefficients indicate association with Parkinson's disease classification, while negative coefficients indicate association with healthy control status. Features are ordered by absolute magnitude of coefficient to highlight the most influential task-feature interactions.

Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized primarily by motor dysfunction, including bradykinesia, rigidity, tremor, and postural instability. It is the second most common neurodegenerative disease worldwide and represents a major source of disability in aging populations. Clinical evaluation of PD motor symptoms is traditionally based on intermittent in-person examinations and ordinal rating scales such as the Unified Parkinson's Disease Rating Scale (Goetz et al., 2008), which are inherently subjective and limited in their ability to capture continuous fluctuations in motor function over time.

Recent advances in wearable sensing technologies have enabled objective, high-frequency measurement of motor behavior in both clinical and free-living environments. Wrist-worn inertial measurement units (IMUs), including accelerometers and gyroscopes embedded in consumer smartwatches, provide a scalable platform for continuous digital phenotyping of movement disorders. These devices can capture subtle alterations in movement speed, amplitude, and rhythmicity that may not be easily detectable during routine clinical assessments, supporting the development of digital biomarkers for neurological disease monitoring (Godinho et al., 2016; Espay et al., 2016).

In parallel, publicly available datasets such as the Parkinson's Disease Smartwatch dataset released on PhysioNet have accelerated research in this area by providing synchronized bilateral wrist sensor recordings during structured motor tasks designed to elicit disease-relevant movement patterns (PhysioNet, 2020). These datasets include both Parkinson's disease patients and healthy controls performing standardized activities under controlled conditions, enabling systematic evaluation of machine learning approaches for motor impairment detection.

Prior work has demonstrated that wearable sensor data can be used to classify Parkinsonian motor states and estimate disease severity using a range of machine learning and deep learning methods, including convolutional neural networks and ensemble approaches. These studies have shown that inertial signals contain discriminative information capable of separating Parkinson's disease from healthy controls, even under substantial inter-subject variability (Hammerla et al., 2015; Zhang et al., 2018). However, many existing approaches prioritize predictive accuracy over interpretability and often rely on complex models that provide limited insight into the underlying physiological or behavioral drivers of classification performance.

Despite these advances, several challenges remain in translating wearable sensor data into clinically meaningful biomarkers. First, raw inertial signals are high-dimensional and noisy, requiring careful preprocessing and feature engineering to extract stable representations. Second, motor impairment in Parkinson's disease is heterogeneous and strongly dependent on behavioral context, with different tasks eliciting distinct movement dynamics. Third, many existing frameworks do not explicitly incorporate task structure, limiting their ability to capture context-dependent variations in motor performance.

To address these limitations, we propose a task-aware computational framework for deriving interpretable motor biomarkers from smartwatch-based accelerometry data in the PhysioNet Parkinson's disease smartwatch dataset (PhysioNet, 2020). Our approach transforms raw bilateral wrist sensor signals into structured representations capturing motor variability, temporal complexity, and movement dynamics across multiple task segments. We further incorporate pseudo-task segmentation to model context-dependent variations in motor behavior and evaluate both group-level separability and predictive performance using interpretable statistical and machine learning methods.

We specifically aim to (i) characterize computational motor features that differentiate Parkinson's disease from healthy controls, (ii) quantify clinically meaningful signatures such as bilateral motor asymmetry, (iii) evaluate the predictive performance of task-aware feature representations using cross-validated classification, and (iv) interpret feature contributions across tasks to identify context-dependent motor signatures of disease. This framework emphasizes interpretability and physiological grounding over model complexity, with the goal of enabling more transparent and clinically actionable wearable-based digital biomarkers.

Methods

We used the PhysioNet Parkinson's Disease Smartwatch dataset, which contains bilateral wrist-worn inertial measurement unit (IMU) recordings collected during standardized motor tasks. The dataset includes participants diagnosed with Parkinson's disease (PD) and healthy controls (HC), each performing structured upper-limb movements designed to elicit a range of motor behaviors. Each recording is associated with a binary label indicating diagnostic class, and all analyses were performed at the subject level using the official preprocessed dataset files.

Raw inertial sensor recordings were loaded from binary files and reshaped into fixed-length multichannel time-series representations consistent with the dataset structure. Each recording was represented as a matrix of shape T×976T \times 976T×976, where TTT denotes the number of temporal samples and 976 corresponds to concatenated sensor-derived channels provided by the dataset preprocessing pipeline. To reduce inter-recording variability and stabilize feature computation, each recording was standardized using z-score normalization, defined as x′=(x−μ)/(σ+ϵ)x' = (x - \mu)/(\sigma + \epsilon)x′=(x−μ)/(σ+ϵ), where μ\muμ and σ\sigmaσ are computed per recording and ϵ\epsilonϵ is a small constant for numerical stability.

For visualization and signal validation purposes, left and right wrist signals were separated based on dataset structure. When required, signals were temporally aligned by truncating to the minimum shared sequence length between limbs to ensure consistent comparison and avoid boundary artifacts. This preprocessing step supported qualitative assessment of raw versus processed signals.

We extracted computational motor features designed to capture variability, temporal dynamics, and signal complexity from the normalized sensor data. These included signal variance as a measure of movement amplitude variability, temporal jerkiness computed as the variance of first-order temporal differences to quantify movement smoothness, and spectral entropy derived from the normalized power spectral density obtained via real-valued fast Fourier transform (FFT), capturing the complexity of motor signals in the frequency domain. These features were computed per channel and averaged to produce subject-level representations.

To quantify bilateral motor impairment, we defined a motor asymmetry index as the difference in signal variance between left and right wrist signals. This metric captures inter-limb imbalance in motor activity and was used for group-level comparisons between PD and HC participants.

To incorporate behavioral structure into the modeling framework, each recording was partitioned into eight equal-length temporal segments approximating structured motor task progression. Although explicit task boundary annotations were not used, this segmentation provides a structured approximation of task-dependent motor behavior. Within each segment, the same set of features (variance, spectral entropy, and jerkiness) was computed. The resulting task-specific features were concatenated to form a structured feature vector encoding context-dependent motor dynamics across the full recording.

For classification, we trained a logistic regression model to distinguish PD from HC participants using the task-aware feature representation. Logistic regression was selected for its interpretability and suitability for high-dimensional structured feature spaces. Model parameters were estimated using standard maximum likelihood optimization.

Model performance was evaluated using stratified five-fold cross-validation to ensure balanced class representation across folds. In each iteration, the model was trained on four folds and evaluated on the held-out fold. Predictive performance was quantified using receiver operating characteristic (ROC) curves and area under the curve (AUC), with final performance reported as the mean AUC and standard deviation across folds.

Prior to model fitting, all features were standardized using z-score normalization to ensure comparability across dimensions and enable meaningful interpretation of logistic regression coefficients. This standardization allows coefficient magnitudes to reflect relative feature importance within the learned decision function.

To interpret the trained model, we analyzed logistic regression coefficients corresponding to each task-feature combination. Positive coefficients indicate association with Parkinson's disease classification, while negative coefficients indicate association with healthy control status. Features were ranked by absolute coefficient magnitude to identify the most influential motor-task interactions contributing to classification.

Group-level statistical comparisons between PD and HC cohorts were performed using the Mann–Whitney U test due to potential non-normality in feature distributions. Statistical significance was assessed for each feature comparison presented in the feature analysis figures.

All analyses were implemented in Python using NumPy, Pandas, Matplotlib, and scikit-learn. Signal processing, feature extraction, model training, and evaluation were performed using a reproducible computational pipeline applied consistently across all subjects.

Results

We evaluated whether smartwatch-derived inertial signals can be transformed into interpretable computational biomarkers of Parkinson's disease through a structured pipeline spanning signal preprocessing, feature engineering, task-aware modeling, and interpretability analysis.

We first examined the transformation of raw wrist-worn accelerometry signals through the preprocessing pipeline (Figure 2). Representative examples from a single subject illustrate the distinction between raw and processed signals for both left and right wrists during a standardized motor task. Raw signals exhibited substantial variability in amplitude, noise, and baseline drift, consistent with the inherent variability of wearable inertial recordings. Following preprocessing, including normalization and temporal alignment, the signals became more stable while preserving underlying tri-axial structure across X, Y, and Z channels. This transformation provided a more consistent representation of motor activity suitable for quantitative feature extraction and downstream modeling.

We next evaluated whether computational features extracted from these processed signals capture meaningful differences between Parkinson's disease and healthy control participants (Figure 3). We derived three complementary motor descriptors—signal variability, spectral entropy, and jerkiness—intended to quantify movement amplitude, temporal complexity, and smoothness of motion, respectively. Across all features, individuals with Parkinson's disease demonstrated statistically significant differences compared to healthy controls. In particular, Parkinson's disease participants exhibited increased signal variability and jerkiness, consistent with more irregular and less smooth motor patterns, as well as altered spectral entropy, reflecting differences in the complexity of movement dynamics. These results indicate that wearable-derived features contain structured information capable of separating disease states at the group level.

We further investigated whether wearable signals capture clinically meaningful lateralized motor impairment characteristic of Parkinson's disease (Figure 4). Using a bilateral asymmetry index defined as the difference in signal variance between left and right wrists, we observed clear group-level differences between Parkinson's disease and healthy control participants. Healthy individuals exhibited relatively symmetric distributions centered near zero, whereas Parkinson's disease participants demonstrated a broader and shifted distribution, indicating increased inter-limb imbalance. This finding is consistent with the known asymmetric onset and progression of motor symptoms in Parkinson's disease and supports the clinical interpretability of the derived computational features.

We then evaluated the predictive capacity of the extracted features using a task-aware classification framework (Figure 5). A logistic regression model trained on concatenated task-segmented features achieved consistent performance across stratified five-fold cross-validation. The model demonstrated a mean area under the receiver operating characteristic curve (AUC) of approximately 0.75, indicating robust discrimination between Parkinson's disease and healthy control participants. The consistency of ROC curves across folds suggests that the learned decision boundary is stable and not driven by a particular subset of the data. Importantly, incorporating task-structured feature representations provided a compact and effective encoding of motor behavior that supported reliable classification performance using a simple linear model.

Finally, we examined the interpretability of the task-aware model by analyzing the learned logistic regression coefficients (Figure 6). Coefficient analysis revealed heterogeneous contributions of motor features across different task contexts, indicating that disease-related motor signatures are not uniformly distributed across behaviors. Certain task-feature combinations contributed more strongly to classification decisions, particularly those capturing variability and movement dynamics under specific motor conditions. This pattern suggests that Parkinson's disease-related motor impairment manifests in a context-dependent manner, with different tasks amplifying or attenuating discriminative motor signatures. The interpretability analysis therefore provides mechanistic insight into how wearable-derived features encode disease-relevant information, highlighting the importance of task structure in computational motor phenotyping.

Together, these results demonstrate that raw smartwatch-based inertial signals can be systematically transformed into interpretable, task-aware computational biomarkers that capture clinically relevant aspects of Parkinsonian motor dysfunction, support disease classification, and provide insight into the behavioral structure underlying model predictions.

Discussion

In this study, we presented a task-aware computational framework for deriving interpretable motor biomarkers of Parkinson's disease (PD) from smartwatch-based inertial sensor data. Using the PhysioNet Parkinson's Disease Smartwatch dataset, we demonstrated that raw bilateral wrist accelerometry signals can be transformed into structured representations that capture clinically meaningful differences between individuals with PD and healthy controls. Across multiple analytical levels—including feature distribution analysis, bilateral asymmetry, classification performance, and model interpretability—our results consistently show that wearable-derived motor signals contain robust and discriminative information about Parkinsonian motor dysfunction.

A key finding of this work is that simple computational descriptors of movement dynamics, including signal variability, spectral entropy, and jerkiness, are sufficient to differentiate PD from healthy controls at the group level. As shown in the feature analysis results, individuals with Parkinson's disease exhibited increased variability and movement irregularity, consistent with impaired motor control and reduced movement smoothness. These findings align with prior literature demonstrating that PD is associated with increased motor noise and disrupted motor execution patterns observable in accelerometry-based measurements.

Beyond global feature differences, our analysis of bilateral motor asymmetry revealed clear group-level divergence between PD and healthy participants. Specifically, PD participants exhibited a broader and more shifted distribution of left-right variability differences, reflecting increased inter-limb motor imbalance. This is consistent with the well-established clinical observation that Parkinson's disease often presents asymmetrically, particularly in early and mid-stage disease progression. Importantly, this result demonstrates that relatively simple signal-derived metrics can capture clinically meaningful characteristics of disease expression without requiring complex modeling approaches.

We further demonstrated that incorporating task structure into feature representations improves both predictive performance and interpretability. The task-aware logistic regression model achieved consistent cross-validated performance (mean AUC $\approx$ 0.75), indicating that structured segmentation of motor behavior provides a stable and informative representation of disease state. While the segmentation used in this work is a uniform temporal partition rather than explicitly annotated task boundaries, the resulting feature structure nevertheless captures context-dependent variations in motor behavior. This suggests that even coarse approximations of behavioral context can enhance the informativeness of wearable-derived features.

Importantly, the interpretability analysis of model coefficients revealed heterogeneous contributions of motor features across task segments. Rather than relying on a single dominant biomarker, the model integrates multiple task-feature interactions, suggesting that Parkinsonian motor impairment is context-dependent and varies across movement conditions. Certain task segments contributed more strongly to classification decisions, particularly those capturing increased variability and movement irregularity. This supports the notion that motor impairment in PD is not static, but dynamically expressed depending on behavioral context.

From a methodological perspective, this work emphasizes the value of interpretable models in wearable digital biomarker research. While more complex deep learning approaches may achieve higher predictive performance in some settings, they often lack transparency and clinical interpretability. In contrast, the linear modeling framework used here enables direct interpretation of feature contributions, facilitating insight into the underlying structure of motor impairment. This trade-off between interpretability and complexity is particularly important in translational clinical applications where understanding model behavior is as important as predictive accuracy.

Several limitations should be acknowledged. First, the task segmentation approach used in this study is an approximation of true behavioral structure and does not rely on explicit task boundary annotations. While this provides a useful framework for structured feature learning, future work incorporating precise task labels or continuous behavioral segmentation may yield more refined representations. Second, the study relies on a controlled dataset, which may not fully capture the variability of real-world free-living conditions. Third, the use of logistic regression, while advantageous for interpretability, may limit the ability to capture nonlinear relationships in the data. Future studies could explore more expressive models while preserving interpretability through techniques such as feature attribution or hybrid modeling approaches.

Despite these limitations, the present findings support the feasibility of deriving clinically interpretable digital motor biomarkers from smartwatch data using relatively simple and transparent computational methods. The consistency of results across feature analysis, asymmetry measures, classification performance, and model interpretability suggests that structured feature engineering combined with task-aware representations provides a robust framework for wearable-based assessment of Parkinson's disease.

Conclusion

This study demonstrates that smartwatch-derived inertial sensor data can be transformed into interpretable computational motor biomarkers capable of distinguishing Parkinson's disease from healthy controls. By integrating signal preprocessing, structured feature extraction, bilateral asymmetry analysis, and task-aware modeling, we show that clinically meaningful patterns of motor dysfunction can be captured using transparent and reproducible computational methods.

The proposed framework achieves consistent classification performance while maintaining interpretability through linear modeling and feature-level analysis. Importantly, our results highlight the value of incorporating behavioral structure into wearable signal analysis, even under simplified task segmentation assumptions. This approach enables both predictive modeling and mechanistic interpretation of motor impairment in Parkinson's disease.

Overall, this work supports the potential of interpretable wearable-based digital phenotyping as a scalable approach for objective assessment of motor function in neurological disorders. Future work should extend this framework to real-world longitudinal data, incorporate finer-grained behavioral segmentation, and evaluate generalizability across independent cohorts.

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